Predicting Diabetes Disease Progression with Regression in Excel

Predictive Analytics Project

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Project type: Predictive Analytics

Data used: Diabetes dataset (442 patients, 10 health variables)

Tool used: Microsoft Excel

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2. Abstract

442	10	0.54	54
patients	health variables	R-squared on test data	typical error (RMSE)

This project predicts how much a diabetes patient's disease will progress in one year. The data has 442 patients. For each patient, ten baseline measurements are given: age, sex, body mass index (BMI), blood pressure and six blood test values.

All the work was done in **Microsoft Excel** using multiple linear regression (the LINEST function). The first 353 patients were used to build the models and the last 89 patients were kept hidden to test them fairly. Four models were compared: a baseline, a simple model with BMI only, a reduced model with three variables and a full model with all ten.

The full model was best. On the hidden test patients it explained about 54% of the differences in progression and its predictions were off by about 54 points on average (RMSE), which is 33% better than guessing the average. The strongest predictors were **s5**, **bmi** and **bp**.

> **In simple words:** We used Excel to find a formula that predicts disease progression from a patient's health numbers. It is much better than a blind guess, but not perfect.

3. Introduction

What is predictive analytics? It uses past data to estimate what will happen for a new case. A bank guesses if a loan will be repaid; a shop guesses next month's sales.

What is regression? When the thing we want to predict is a number (such as sales, price or a score), the task is called regression. The result is a formula like: predicted value = a + b1 x variable 1 + b2 x variable 2 + ...

Why Excel? Excel can fit regression models with the LINEST function and check them with simple formulas, so no programming is needed. This makes the method easy to understand and to audit.

Objectives

- Build regression models in Excel to predict disease progression.
- Test them on patients the model has never seen.
- Compare simple and complex models.
- Find which health variables matter most.
- Understand how the same idea is used in finance.

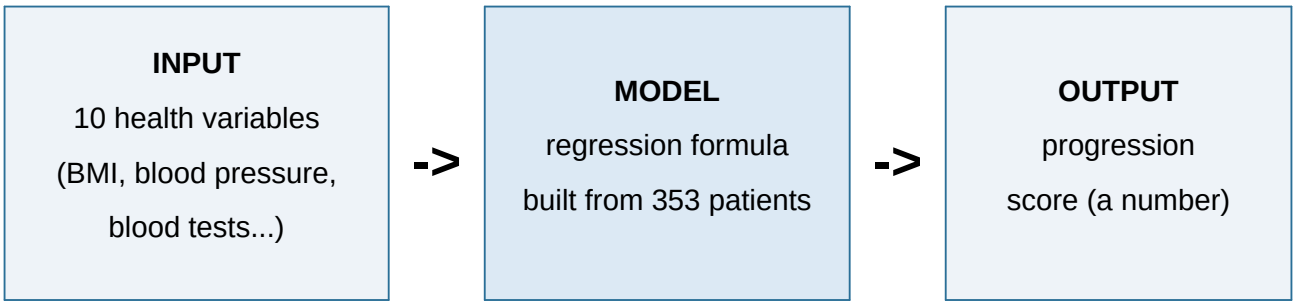
> **In simple words:** The project is like learning a rule from 353 solved examples and then checking it on 89 new cases.

4. Problem Statement

Doctors want to know early which patients are likely to get worse, so that they can give extra care. Measuring progression after a year is slow. Can we predict it from the first-visit measurements?

Problem: Predict the one-year disease progression score of a patient from ten baseline health variables.

How the model works



What counts as success?

- **Low error (RMSE and MAE):** predictions should be close to the real scores.
- **High R-squared:** the model should explain a large share of the differences between patients.
- **Beats the baseline:** it must be better than always guessing the average.

In simple words: A model is useful only if it predicts better than a simple guess.

5. Data Discussion

Source: The Diabetes dataset (Efron et al., 2004), a real medical dataset available in scikit-learn. **Size:** 442 patients and 10 baseline variables. **Target:** disease progression one year later (a score from 25 to 346).

Variable	What it means in plain words
bmi	Body mass index (weight relative to height)
bp	Average blood pressure
s5	Log of serum triglycerides (a blood fat)
age, sex	Age in years and sex code
s1, s2, s3, s4	Cholesterol-related blood tests (total, LDL, HDL, ratio)
s6	Blood sugar level
Progression	Disease progression after one year (the value to predict)

Excel data (sample of 8 rows)

ID	bmi	bp	s5	age	Progression	Set	Predicted	Error
1	32.1	101	4.86	59	151	Train	203	-52
2	21.6	87	3.89	48	75	Train	69	6
3	30.5	93	4.67	72	141	Train	175	-34
4	25.3	84	4.89	24	206	Train	167	39
5	23.0	101	4.29	50	135	Train	127	8
6	22.6	89	4.19	23	97	Train	106	-9
7	22.0	90	3.95	36	138	Train	75	63
8	26.2	114	4.25	66	63	Train	117	-54

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Data (442 rows; Predicted and Error columns are Excel formulas)

In simple words: There are 0 missing values. The first 353 rows are used to build the model (Train) and the last 89 to test it (Test).

6. Exploratory Analysis

We first look at the data to understand it.

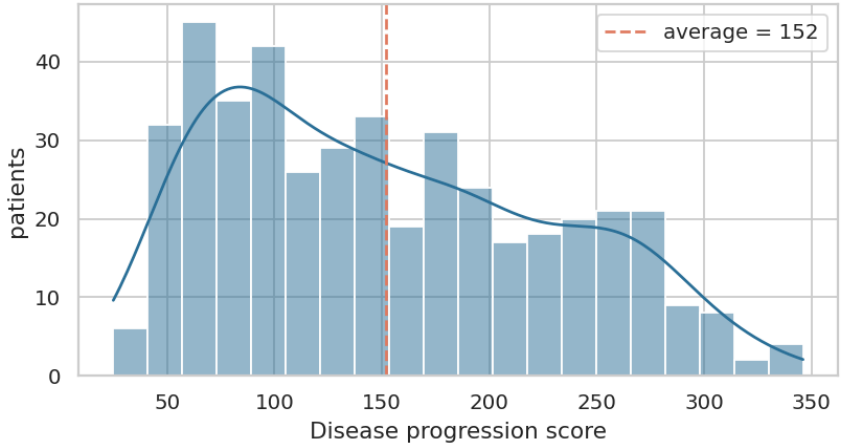


Figure 1: Distribution of the disease progression score

Most patients have a progression score between 50 and 250 (average 152). A few patients have very high scores.

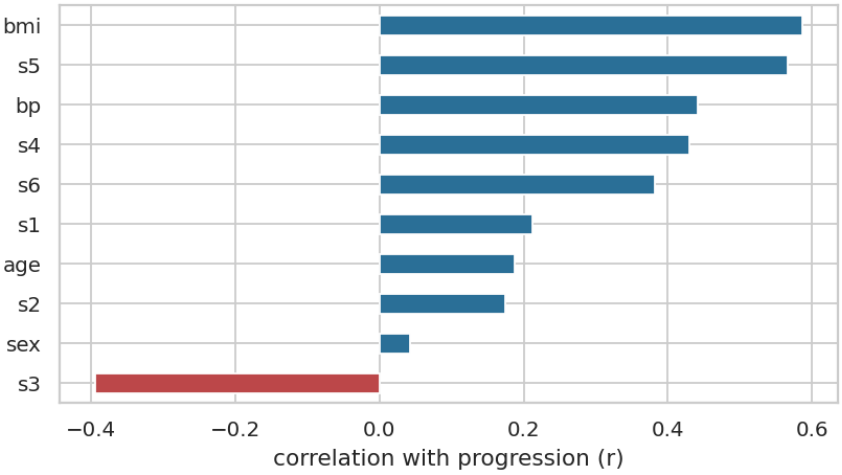


Figure 2: Correlation of each variable with progression (blue = moves together, red = moves opposite)

The strongest links are **bmi** (r = 0.59) and **s5** (r = 0.57), then **bp** (r = 0.44). Higher BMI, blood pressure and blood fats go with faster progression.

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Correlation_With_Progression

7. Method, Tools and Technologies

Tool used

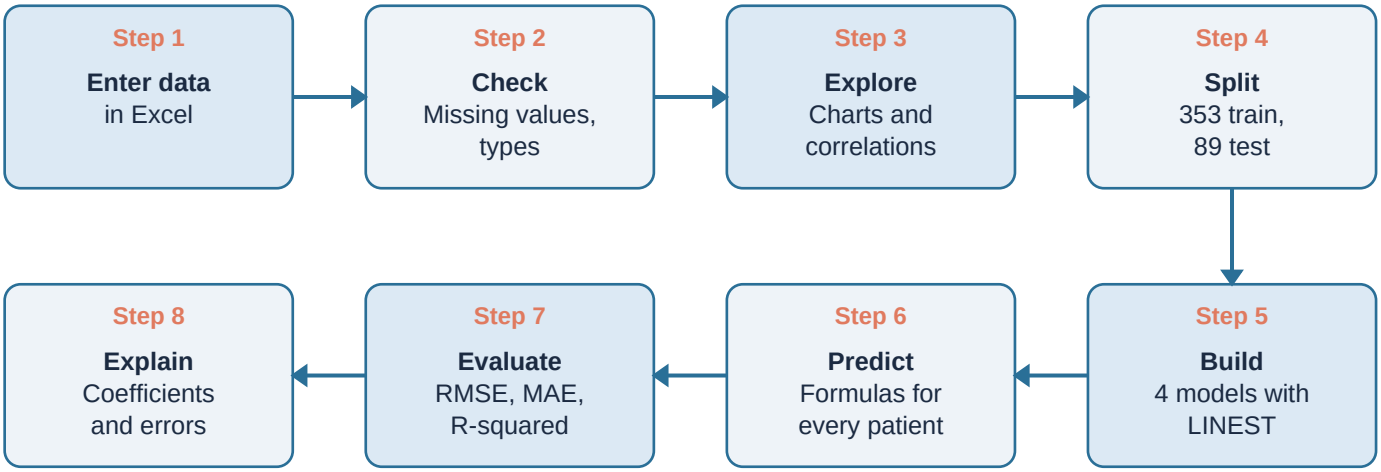
Tool	Used for
Microsoft Excel	Data handling, regression with the LINEST function, predictions, error measures and tables, all with formulas. The charts in this report were drawn from the Excel results.

Methods used

Method	How it works in simple words
Multiple linear regression	Finds the best straight-line formula linking the variables to the result
Train-test split	Build the model on 353 patients, test it on 89 hidden ones
RMSE and MAE	Average size of the prediction mistakes (lower is better)
R-squared	Share of the differences between patients explained by the model (higher is better)
t-test and p-value	Checks if a variable really matters
Standardised coefficient	Compares the importance of variables on the same scale

8. Workflow

The project followed these steps in order.



What happens at each step

- **Split:** hide 89 patients so the final test is fair.
- **Build:** fit the baseline, simple, reduced and full models on the training rows only.
- **Predict and evaluate:** apply each formula to every row and compare with the real score.
- **Explain:** read the coefficients to see which variables matter and where errors are large.

> **In simple words:** Study on the training rows, then take the exam on the hidden rows.

9. Procedure

Step	Excel formulas used	Sheet
Build models	INDEX(LINEST(y, x, TRUE, TRUE), row, column)	Models
Predictions	intercept + SUMPRODUCT(x values, coefficients)	Data
Baseline	AVERAGE of the training scores	Models
Error measures	SQRT(SUMXMY2()/n), SUMPRODUCT(ABS()), 1 - SUMXMY2/DEVSQ	Evaluation
Importance	coefficient / std error, TDIST, STDEV, RANK	Coefficients
Error study	AVERAGEIFS, COUNTIFS, MAXIFS	Error_Analysis
Correlation	CORREL	Correlation_With_Progression

1. Entered 442 rows and marked the first 353 as Train and the last 89 as Test.
2. Fitted four models on the training rows only: baseline (average), simple (bmi), reduced (bmi, bp, s5) and full (all 10 variables).
3. Wrote one prediction formula per patient for every model.
4. Calculated RMSE, MAE and R-squared separately for train and test rows.
5. Read the full model's coefficients, t-values and p-values.
6. Studied the errors by progression level and checked how many predictions were close.

In simple words: Every number comes from a formula, so the workbook can be checked cell by cell.

10. Results: Model Comparison

All models were scored on the same 89 hidden test patients.

Model	Train RMSE	Test RMSE	Test MAE	Test R2	Better than baseline
Baseline (average)	76.2	80.2	70.2	-0.002	-
Simple (bmi)	62.0	63.9	53.8	0.365	20.4%
Reduced (bmi, bp, s5)	55.4	56.1	46.2	0.509	30.0%
Full (10 variables)	53.4	54.1	42.5	0.544	32.5%

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Evaluation

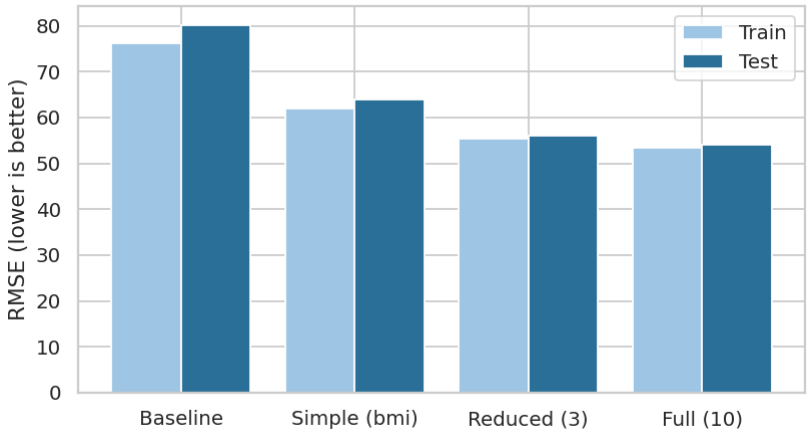


Figure 3: RMSE of each model on training and test patients (lower is better)

The **full model** is best: test RMSE 54.1, MAE 42.5 and R-squared 0.544. Guessing the average gives RMSE 80.2. So the model cuts the typical error by about 33%. Adding variables helped each time, but the gain from 3 to 10 variables is small.

Meaning of the scores

- **RMSE / MAE:** typical size of the mistake, in progression points.
- **R-squared:** 0 means no better than the average; 1 means perfect.
- Train and test scores are close, so the model is not just memorising.

11. Actual vs Predicted and Residuals

If the model were perfect, every dot in the left chart would lie on the dashed line. The right chart shows the errors (actual minus predicted).

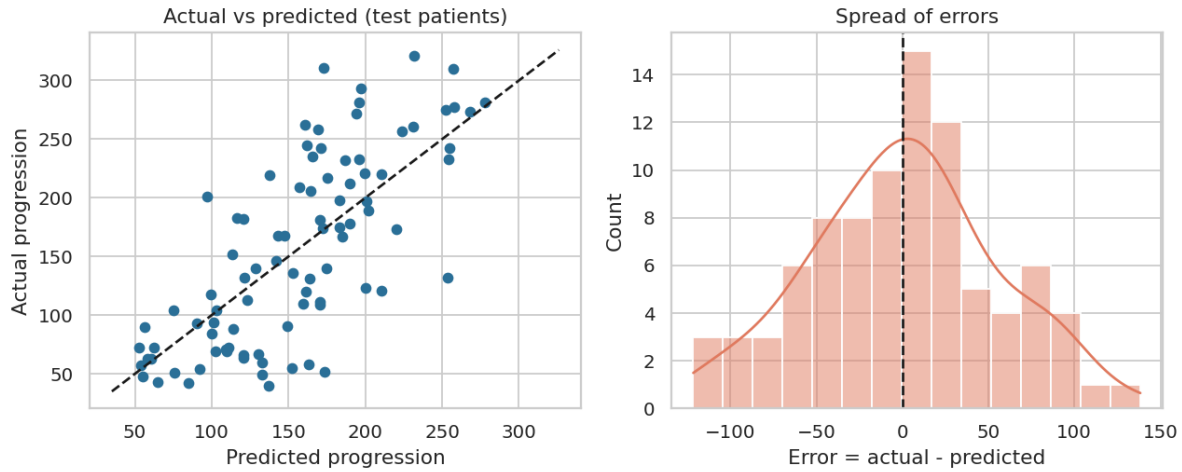


Figure 4: Actual vs predicted scores and the spread of errors for the 89 test patients

The dots follow the line but are scattered around it, which fits an R-squared of 0.54. The errors are centred near zero (average error 0.4), so the model is not biased up or down overall. Errors are larger for patients with very high scores.

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Data (Pred_Full and Error_Full columns)

In simple words: The model gets the general direction right, but individual predictions can be off by 40 to 60 points.

12. Feature Importance (Coefficients)

The coefficient tells how many progression points change when a variable rises by one unit. The p-value tells if the effect is real (below 0.05). The standardised coefficient lets us compare variables fairly.

Variable	Coefficient	t-value	p-value	Matters?	Std. coef.
s5	58.258	3.25	0.001	Yes	0.39
bmi	5.538	6.61	< 0.001	Yes	0.31
s1	-0.654	-0.96	0.339	No	-0.29
bp	1.011	4.03	< 0.001	Yes	0.18
sex	-23.721	-3.60	< 0.001	Yes	-0.16
s2	0.334	0.53	0.598	No	0.13
s4	5.431	0.83	0.407	No	0.09
s6	0.358	1.13	0.258	No	0.05
s3	-0.124	-0.14	0.892	No	-0.02
age	-0.029	-0.12	0.902	No	-0.01

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Coefficients

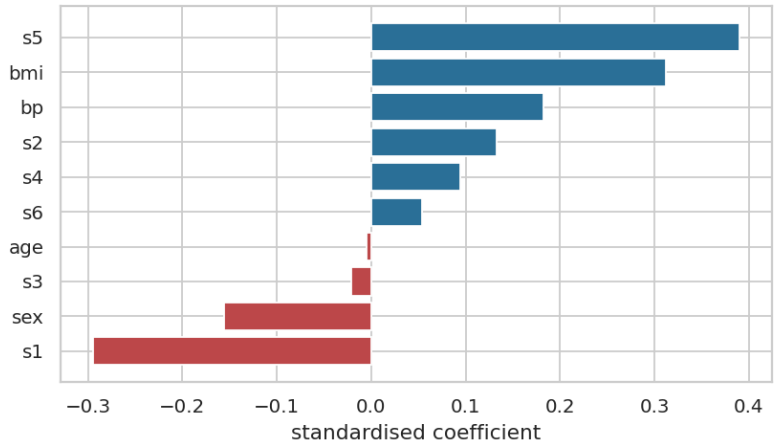


Figure 5: Standardised coefficients (importance) of the full model

4 of 10 variables are statistically significant: bmi, bp, s5, sex. The most important are **s5**, **bmi** and **bp**. Variables such as age, s2 and s3 add almost nothing once the others are in the model.

13. Error Analysis

43	56%	72%	138
average error (MAE)	within 40 points	within 60 points	largest error

On the 89 test patients, the average miss was 43 points. 56% of predictions were within 40 points and 72% within 60. 18 predictions were off by more than 75 points.

Errors by progression level

Actual progression	Patients	Mean absolute error	Average error (bias)
Low (below 100)	29	42.1	-36.9
Medium (100 to 199)	31	32.6	-12.4
High (200 to 299)	26	49.0	46.4
Very high (300 and above)	3	93.5	93.5

Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx | Sheet: Error_Analysis

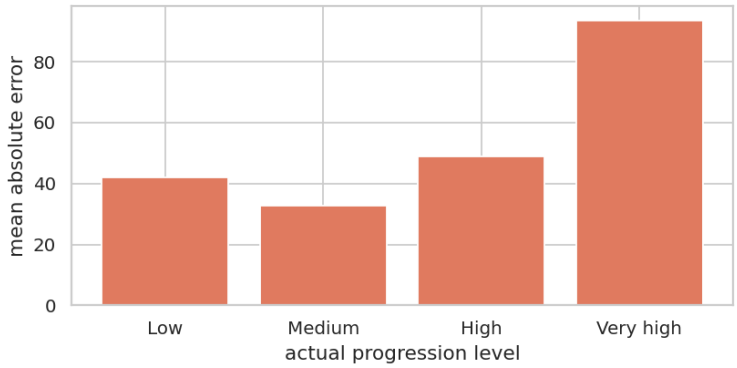


Figure 6: Average error by actual progression level

The model **over-predicts** patients with low progression and **under-predicts** those with very high progression (negative and positive bias). This is common: regression pulls predictions towards the average.

14. Insights (Observations)

1. The full model explains about 54% of the differences in progression on new patients and cuts the typical error by 33% compared with guessing the average.
2. Even a one-variable model (BMI only) is much better than the baseline.
3. Using three variables (bmi, bp, s5) gets most of the benefit: test RMSE 56.1 against 54.1 for all ten.
4. The most important variables are s5, bmi and bp. Higher BMI and blood fats go with faster progression.
5. Train and test errors are similar, so the model works on new patients.
6. Predictions are least reliable for very high or very low progression.
7. About 44% of predictions are more than 40 points away, so the model should support, not replace, a doctor.

In simple words: Good for spotting the general level of risk, not for exact numbers.

15. Relevance to Finance

This project is about health, but regression is one of the most used tools in finance. Only the data changes.

In this project	In finance
Patient	Customer, loan, company or month
Health variables	Income, age, credit score, interest rate, sales history
Progression score	Loan loss amount, sales, share price, insurance claim
RMSE / MAE	Average forecast error in rupees
R-squared	How much of the changes the model explains
Baseline (average)	Simple forecast such as last year's average
Train and test split	Backtesting on data the model has not seen
Coefficient	Effect of one factor, such as the effect of interest rate on demand

Where is it used?

- Sales and demand forecasting
- Estimating loan loss and credit exposure
- Pricing of houses, insurance and loans
- Budgeting and cost forecasting

In simple words: Replace the patient data with company data and the same Excel method predicts sales or costs.

16. Limitations

- **Moderate accuracy:** R-squared of about 0.54 means almost half of the differences remain unexplained.
- **Straight-line assumption:** regression assumes simple linear effects, which may not be true.
- **Small test set:** only 89 test patients, so scores are approximate.
- **One dataset:** 442 patients from one study; the model may not work elsewhere.
- **Linked variables:** the cholesterol variables are related to each other (s1 to s4), which makes single coefficients hard to read.
- **Not medical advice:** the model cannot replace a doctor.
- **Not a finance dataset:** finance relevance is shown by analogy.

In simple words: The model is a useful first step, but it should be checked on more data before real use.

17. Recommendations

1. **Use the model as a helper.** Doctors (or analysts) should make the final decision.
2. **Use the simple 3-variable model** (bmi, bp, s5) when fewer measurements are available. It performs almost as well.
3. **Focus on BMI and blood fats** in early screening, since they matter most.
4. **Try better models** such as decision trees or random forests, which can capture non-linear effects.
5. **Collect more patients** and test on a new group.
6. **Monitor the model** and rebuild it when new data arrives.
7. **For finance:** repeat the same Excel steps on sales or loan-loss data.

18. Conclusion

This project built four regression models in Microsoft Excel to predict one-year diabetes progression. The full model explained about 54% of the differences between unseen patients and cut the typical prediction error by 33% compared with guessing the average. BMI, s5 (blood fats) and blood pressure are the most useful predictors.

The project also showed that a small model can perform almost as well as a large one, and that checking errors by group is as important as looking at one overall score. The same Excel method can be used in finance to forecast sales, costs or loan losses.

What I learned

- How to split data for a fair test.
- How to build and read regression models using LINEST.
- How to measure errors with RMSE, MAE and R-squared.
- How to link a technical result to a practical decision.

19. References

1. Efron, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004). Least Angle Regression. Annals of Statistics, 32(2), 407-499.
2. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12 (source of the dataset copy).
3. Hastie, T., Tibshirani, R. and Friedman, J. (2009). The Elements of Statistical Learning. Springer.
4. Montgomery, D. C., Peck, E. A. and Vining, G. G. Introduction to Linear Regression Analysis. Wiley.
5. Levine, D. M. et al. Statistics for Managers Using Microsoft Excel. Pearson.
6. Microsoft. LINEST function, Excel support documentation.

In simple words: These are the sources for the dataset, the regression method and the Excel functions used.

20. Appendix

A. Sheets in the Excel file: Sujal_Predictive_Diabetes_Analysis.xlsx

Sheet	What it contains
Data	442 patients, Train/Test label, predictions and errors (formulas)
Models	LINEST regression models: baseline, simple, reduced, full
Evaluation	RMSE, MAE and R-squared for train and test
Coefficients	Coefficients, t-values, p-values, standardised coefficients
Error_Analysis	Error measures and errors by progression level
Correlation_With_Progression	Correlation of each variable with the target
Data_Dictionary	Meaning of every column

B. Full model equation (Excel LINEST result)

Predicted progression = -287.2 + 5.538 x bmi + 1.011 x bp + 58.258 x s5 + (seven smaller terms for age, sex, s1, s2, s3, s4 and s6).

C. Short glossary

Term	Simple meaning
Regression	Predicting a number from other numbers
Train / Test	Rows used to build / rows hidden to check the model
RMSE	Typical size of the prediction mistakes
R-squared	Share of differences explained by the model (0 to 1)
p-value	Below 0.05 means the effect is probably real

Tool used: Microsoft Excel. Files: Sujal_Predictive_Analytics_Report.pdf and Sujal_Predictive_Diabetes_Analysis.xlsx.